Jack Vacek
Jaesun Park

Abstract

Medical image classification plays a crucial role with the diagnosis of diseases and within deep learning models, an imbalanced dataset significantly affects the accuracy in evaluation. This project seeks to combat the imbalance's effect through techniques learned from our 421 class - focusing specifically on the AUC, or "Area Under the Curve". By utilizing the MedMNIST dataset, BreastMNIST and PneumoniaMNIST, we trained a total of four different ResNet18 models where two utilized Cross Entropy loss on the two datasets and the others AUCM from the libAUC library. Additionally, we explored techniques to further improve the AUCM loss performance. Through testing the number of epochs, different neural network structures, and parameters of the optimizer, we were able to increase the performance about 14% for the BreastMNIST data and about 0.5% performance increase for the PneumoniaMNIST. The results suggest that strategic adjustments to the number of epochs, the type of model, and learning rate/batch size of the optimizer can lead to significant improvements of the AUCM loss scores, and thus a more accurate model.

Introduction

In the classification of medical images, the accuracy of correctly identifying images is vital to the survival of the patient. However, in medical imaging, diseases tend to be rare within the population, so there are many images of healthy patients and few of ill patients. This leads to an imbalance of the types of images given to the deep learning model to train on. Thus, one might see that the model was able to correctly determine 999 out of 1000 of the healthy images, but only 1 out of 10 of the images demonstrating the disease and with a traditional way of measuring accuracy say that the model was 99% accurate. However, in the medical field this is dangerously misleading. To combat this hazardous misrepresentation, our paper describes our efforts in maximizing the accuracy of the area under the curve of an ROC curve to provide a more telling measure of accuracy. In determining the area under the curve, AUC, one is able to discern how accurate the model is in relation to the true positive and true negative rates. Maximizing this AUC means the model is more accurate in both correctly identifying truly healthy patients and truly ill patients. There are many different ways to maximize the area under the curve; however, our group decided to test how changing the parameters of the optimizer, changing the number of epochs, and even testing different deep learning models would affect the area. In order to obtain a baseline measure of the model's AUC measurement, we trained a ResNet18 model with parameters predetermined by our instructor and compared our results. We decided to test between two additional deep model architectures: GoogleNet and ResNet101 to see if changing the architecture would have an effect on the AUC. As well, as tasked to do so by the instructor, we trained ResNet18 models and measured the area under the curve while using Cross Entropy loss as the loss function.

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Method

Data Description

The project consists of two datasets from MedMNIST. BreastMNIST, which consists of images for breast cancer screening and PenumoniaMNIST, which consists of chest X-rays for pneumonia detection. Both of these datasets are known for their imbalance, making them suitable for our attempts to improve the AUCM loss performance.

Method 1 - Number of epochs

Changing the number of epochs is another important method practiced in order to improve the performance of the model. When choosing the number of epochs to utilize, one must consider that a large number of epochs may lead to overfitting whereas a low number of epochs may lead to risks of underfitting. Hence it is important to decide the number of epochs to utilize as that can determine the overall performance of the model. For more clarity, here is a general rule of thumb with choosing the number of epochs

Low Number of Epochs

- Faster training time and less resources, but can be at risk of underfitting.

High Number of Epochs

- Slower training time and more intensive, but can be at risk of overfitting.

Typically, the solution to this issue is to either perform what is known as “early stopping” or monitor the optimal number of epochs to perform to get the best performance. Early stopping is a form of practice where one stops the training whenever the model performs worse on validation data to minimize the effects of overfitting. Monitoring refers to simply selecting or “cherry picking” the epochs that allow the performance of the model to be at its highest.

Method 2 - Optimizer parameter tuning

The first method we employed on our model was to test different parameters of the optimizer. Each model was trained on a different combination of “learning rates” and “batch size”, in order to find the best combination of parameters that results in the best AUCM loss performance.

Learning Rate

The learning rate is a crucial parameter in training neural networks as it controls or determines how much the model should step per iteration as it moves “toward a minimum of a loss function”. In other words, choosing the right learning rate is crucial for a good training performance as **too low** can make the training process extremely slow, while **too high** can result in the training to go back and forth around the minimum, at a risk of never reaching the minimum.

Training Batch Size

The batch size parameter is a crucial factor that influences the training process of neural networks. It defines the number of training samples to work through before the inner parameters are updated. Typically, a **large batch size** can lead to faster training as it processes more data in a singular "batch" while a **lower batch size** can lead to computationally intensive training. As a lower batch size requires more frequent training compared to a higher batch size.

It is important that each model is to be trained multiple times with different combinations of parameters, to maximize the AUCM loss performance.

Method 3 - Testing different Neural Network structures

In order to further enhance the performance of the model, we explored various types of neural network structures. The primary goal here was to see how much of an influence the type of a model can have on the overall AUCM loss performance. For this experiment, we chose to work with GoogleNet and ResNet101. In order to maximize the accuracy of the results, for each neural network architecture, we also chose the best combination of parameters between learning rate and training batch size. That way, we can see a clear correlation between the overall AUCM loss performance and the neural network structure.

Experiments

Firstly, in our experiments, we wanted to test what the Cross Entropy Loss and the AUCM Loss would be for both the PneumoniaMNIST and BreastMNIST datasets given default parameters. We also were testing to see if changing only the number of epochs would change the AUC score. In doing so, we set the number of epochs to be 20 - at first we tried 100 epochs but the runtime took too long, the learning rate to be 0.1, and the margin to be 1. Then we found the best score and the corresponding epoch in which that score was achieved.
The results for the Cross Entropy Loss score for both datasets are as follows:

	1st Epoch CE Score	Best Epoch CE Score
BreastMNIST	0.3538011695906433	0.6558061821219716
PneumoniaMNIST	0.9100482138943676	0.9702717510409817

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```
!python3 -W ignore demo.py --data=breastmnist --task_index=0 --pos_class=0 --epochs=20 --loss='CE' --model_name="resnet18"
```

```
Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
Epoch:0 Test AUC: 0.3538011695906433
Epoch:1 Test AUC: 0.35192147034252297
Epoch:2 Test AUC: 0.6558061821219716
Epoch:3 Test AUC: 0.4901837928153717
Epoch:4 Test AUC: 0.3312447786131996
Epoch:5 Test AUC: 0.5925229741019215
Epoch:6 Test AUC: 0.6259398496240601
Epoch:7 Test AUC: 0.5380116959064327
Epoch:8 Test AUC: 0.5720551378446115
Epoch:9 Test AUC: 0.5927318295739348
Epoch:10 Test AUC: 0.5670426065162907
Epoch:11 Test AUC: 0.6409774436090225
Epoch:12 Test AUC: 0.5779030910609858
Epoch:13 Test AUC: 0.5964912280701754
Epoch:14 Test AUC: 0.559732664995823
Epoch:15 Test AUC: 0.4273182957393483
Epoch:16 Test AUC: 0.4425647451963241
Epoch:17 Test AUC: 0.5446950710108605
Epoch:18 Test AUC: 0.4540517961570593
Epoch:19 Test AUC: 0.3888888888888889
for model: resnet18 lr: 0.1 batch size: 32 best_epoch: 2 score: 0.6558061821219716
```

```
!python3 -W ignore demo.py --data=pneumoniamnist --task_index=0 --pos_class=0 --epochs=20 --loss='CE' --model_name="resnet18"
```

```
Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
Epoch:0 Test AUC: 0.9100482138943676
Epoch:1 Test AUC: 0.7737015121630508
Epoch:2 Test AUC: 0.9490466798159106
Epoch:3 Test AUC: 0.8896778435239973
Epoch:4 Test AUC: 0.9523668639053255
Epoch:5 Test AUC: 0.9625794433486741
Epoch:6 Test AUC: 0.9640477755862371
Epoch:7 Test AUC: 0.9618233618233618
Epoch:8 Test AUC: 0.9379684418145956
Epoch:9 Test AUC: 0.9007451238220467
Epoch:10 Test AUC: 0.4661078238001315
Epoch:11 Test AUC: 0.9663927240850317
Epoch:12 Test AUC: 0.7867959675651985
Epoch:13 Test AUC: 0.9542735042735043
Epoch:14 Test AUC: 0.9646175761560377
Epoch:15 Test AUC: 0.967367959675652
Epoch:16 Test AUC: 0.9700964277887355
Epoch:17 Test AUC: 0.9623274161735701
Epoch:18 Test AUC: 0.9542625465702389
Epoch:19 Test AUC: 0.9702717510409817
for model: resnet18 lr: 0.1 batch size: 32 best_epoch: 19 score: 0.9702717510409817
```

The results for the AUCM Loss score for both datasets are as follows:

	1st Epoch AUCM Score	Best Epoch AUCM Score
BreastMNIST	0.5812447786131997	0.706140350877193
PneumoniaMNIST	0.9164584703046242	0.9648805610344072

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```
!python3 -W ignore demo.py --data=breastmnist --task_index=0 --pos_class=0 --epochs=20 --loss='AUCM' --model_name="resnet18"
```

```
Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
Epoch:0 Test AUC: 0.5812447786131997
Epoch:1 Test AUC: 0.6205096073517126
Epoch:2 Test AUC: 0.41604010025062654
Epoch:3 Test AUC: 0.5442773600668338
Epoch:4 Test AUC: 0.40183792815371766
Epoch:5 Test AUC: 0.5983709273182958
Epoch:6 Test AUC: 0.5795739348370927
Epoch:7 Test AUC: 0.6712614870509608
Epoch:8 Test AUC: 0.3696741854636591
Epoch:9 Test AUC: 0.5977443609022557
Epoch:10 Test AUC: 0.6489139515455307
Epoch:11 Test AUC: 0.3680033416875522
Epoch:12 Test AUC: 0.6148705096073517
Epoch:13 Test AUC: 0.36612364243943185
Epoch:14 Test AUC: 0.6783625730994152
Epoch:15 Test AUC: 0.692564745196324
Epoch:16 Test AUC: 0.706140350877193
Epoch:17 Test AUC: 0.6058897243107769
Epoch:18 Test AUC: 0.4076858813700919
Epoch:19 Test AUC: 0.56265664160401
for model: resnet18 lr: 0.1 batch size: 32 best_epoch: 16 score: 0.706140350877193
```

```
!python3 -W ignore demo.py --data=pneumoniamnist --task_index=0 --pos_class=0 --epochs=20 --loss='AUCM' --model_name="resnet18"
```

```
Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
Epoch:0 Test AUC: 0.9164584703046242
Epoch:1 Test AUC: 0.9126780626780626
Epoch:2 Test AUC: 0.937869822485207
Epoch:3 Test AUC: 0.9011505588428665
Epoch:4 Test AUC: 0.9466688582073198
Epoch:5 Test AUC: 0.9310650887573965
Epoch:6 Test AUC: 0.9456278763971071
Epoch:7 Test AUC: 0.5840346263423186
Epoch:8 Test AUC: 0.9580100810870041
Epoch:9 Test AUC: 0.9648805610344072
Epoch:10 Test AUC: 0.9492548761779531
Epoch:11 Test AUC: 0.949200087661626
Epoch:12 Test AUC: 0.9118343195266273
Epoch:13 Test AUC: 0.9497698882314267
Epoch:14 Test AUC: 0.9522572868726714
Epoch:15 Test AUC: 0.9393710278325663
Epoch:16 Test AUC: 0.9273613850536927
Epoch:17 Test AUC: 0.9562458908612754
Epoch:18 Test AUC: 0.9207977207977208
Epoch:19 Test AUC: 0.9150120534735918
for model: resnet18 lr: 0.1 batch size: 32 best_epoch: 9 score: 0.9648805610344072
```

Previously stated in our methods section, the first experiment we took to optimize the AUC performance was through testing different numbers of epochs, which is the number of times the data is used to train the model. Our experiment wielded very informative results. Changing the number of epochs can help train the model to become more accurate as it runs through the data more or could make the model overfit by running the data too much through the model. Thus, we kept track of the original model's score when the training data was run through once and the best score and epoch. As we can see from the screenshots above, most of the

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models followed the general idea stated in the methods section above. The models initially became generally more accurate as the number of epochs increased; however, after a certain threshold, they started to become generally less accurate. The only exception is when testing the CE Loss for the pneumoniaMNIST dataset, which possibly could have been further trained to become more accurate; however, this was too computationally expensive to compute. These results mirror the theory behind early stopping, as one would stop training the model by running more epochs as the validation error in the model started to increase.

In discovering this idea, we decided to apply the same principle to the two additional neural networks, ResNet101 and GoogleNet, when finding the most efficient model. However, in addition to testing the number of epochs in these two additional models in the same fashion as we did in ResNet18, we also wanted to see how changing the neural network's optimizer parameters would affect the overall performance for every model. We decided to examine how changing the learning rate, between 0.01, 0.1, 0.5, and 1, and batch size, between 16, 32, and 64, would affect each model's performance. Below are the results of our experiments when running these different optimizer parameters on each model in tandem with finding the epoch that gives the best performance of the 20 on each dataset.

For the ResNet18 on BreastMNIST:

```
Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 0.01 batch size: 16 best_epoch: 16 score: 0.8076441102756893

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 0.01 batch size: 32 best_epoch: 16 score: 0.7502088554720133

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 0.01 batch size: 64 best_epoch: 19 score: 0.8395989974937343

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 0.1 batch size: 16 best_epoch: 19 score: 0.7412280701754386

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 0.1 batch size: 32 best_epoch: 16 score: 0.706140350877193

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 0.1 batch size: 64 best_epoch: 19 score: 0.8460735171261486

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 0.5 batch size: 16 best_epoch: 19 score: 0.7395572263993317
```

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Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 0.5 batch size: 32 best_epoch: 16 score: 0.5668337510442774

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 0.5 batch size: 64 best_epoch: 0 score: 0.4692982456140351

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 1.0 batch size: 16 best_epoch: 19 score: 0.6977861319966583

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 1.0 batch size: 32 best_epoch: 6 score: 0.4578111946532999

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet18
for model: resnet18 lr: 1.0 batch size: 64 best_epoch: 3 score: 0.5394736842105263

For ResNet18 on PneumoniaMNIST:

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 0.01 batch size: 16 best_epoch: 4 score: 0.9500328731097961

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 0.01 batch size: 32 best_epoch: 10 score: 0.9504163927240851

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 0.01 batch size: 64 best_epoch: 11 score: 0.9432610124917817

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 0.1 batch size: 16 best_epoch: 14 score: 0.9700964277887355

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 0.1 batch size: 32 best_epoch: 9 score: 0.9648805610344072

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 0.1 batch size: 64 best_epoch: 6 score: 0.955698005698006

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 0.5 batch size: 16 best_epoch: 15 score: 0.9661078238001315

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Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 0.5 batch size: 32 best_epoch: 16 score: 0.9672145518299364

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 0.5 batch size: 64 best_epoch: 15 score: 0.9577909270216963

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 1.0 batch size: 16 best_epoch: 14 score: 0.950668419899189

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 1.0 batch size: 32 best_epoch: 5 score: 0.9649901380670611

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet18
for model: resnet18 lr: 1.0 batch size: 64 best_epoch: 16 score: 0.966787201402586

For ResNet101 on BreastMNIST:

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 0.01 batch size: 16 best_epoch: 9 score: 0.6785714285714286

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 0.01 batch size: 32 best_epoch: 9 score: 0.5952380952380952

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 0.01 batch size: 64 best_epoch: 13 score: 0.5378028404344194

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 0.1 batch size: 16 best_epoch: 14 score: 0.599624060150376

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 0.1 batch size: 32 best_epoch: 11 score: 0.5864661654135339

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 0.1 batch size: 64 best_epoch: 2 score: 0.5653717627401837

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 0.5 batch size: 16 best_epoch: 16 score: 0.5693400167084377

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Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 0.5 batch size: 32 best_epoch: 18 score: 0.6063074352548037

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 0.5 batch size: 64 best_epoch: 13 score: 0.525062656641604

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 1.0 batch size: 16 best_epoch: 10 score: 0.6000417710944026

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 1.0 batch size: 32 best_epoch: 15 score: 0.6558061821219716

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: resnet101
for model: resnet101 lr: 1.0 batch size: 64 best_epoch: 18 score: 0.5025062656641603

For ResNet101 on PneumoniaMNIST:

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 0.01 batch size: 16 best_epoch: 19 score: 0.9367959675651983

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 0.01 batch size: 32 best_epoch: 15 score: 0.9275367083059392

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 0.01 batch size: 64 best_epoch: 16 score: 0.3956059609905764

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 0.1 batch size: 16 best_epoch: 18 score: 0.9599386368617138

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 0.1 batch size: 32 best_epoch: 19 score: 0.9071115494192417

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 0.1 batch size: 64 best_epoch: 19 score: 0.2648367302213456

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 0.5 batch size: 16 best_epoch: 19 score: 0.9380013149243918

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Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 0.5 batch size: 32 best_epoch: 8 score: 0.9435349550734166

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 0.5 batch size: 64 best_epoch: 19 score: 0.9366425597194828

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 1.0 batch size: 16 best_epoch: 13 score: 0.9421542844619769

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 1.0 batch size: 32 best_epoch: 13 score: 0.9436664475126014

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: resnet101
for model: resnet101 lr: 1.0 batch size: 64 best_epoch: 17 score: 0.9503506465044927

For GoogleNet on BreastMNIST:

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 0.01 batch size: 16 best_epoch: 18 score: 0.8256056808688388

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 0.01 batch size: 32 best_epoch: 9 score: 0.815998329156224

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 0.01 batch size: 64 best_epoch: 17 score: 0.8293650793650794

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 0.1 batch size: 16 best_epoch: 19 score: 0.8771929824561403

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 0.1 batch size: 32 best_epoch: 13 score: 0.8830409356725145

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 0.1 batch size: 64 best_epoch: 9 score: 0.8613199665831245

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 0.5 batch size: 16 best_epoch: 19 score: 0.7919799498746867

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Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 0.5 batch size: 32 best_epoch: 18 score: 0.7715121136173768

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 0.5 batch size: 64 best_epoch: 15 score: 0.8663324979114453

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 1.0 batch size: 16 best_epoch: 17 score: 0.6727234753550543

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 1.0 batch size: 32 best_epoch: 19 score: 0.6593567251461989

Using downloaded and verified file: ./med/breastmnist.npz
Using downloaded and verified file: ./med/breastmnist.npz
model_name: googlenet
for model: googlenet lr: 1.0 batch size: 64 best_epoch: 19 score: 0.6787802840434419

For GoogleNet on PneumoniaMNIST:

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 0.01 batch size: 16 best_epoch: 10 score: 0.9671378479070786

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 0.01 batch size: 32 best_epoch: 15 score: 0.9627328511943897

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 0.01 batch size: 64 best_epoch: 18 score: 0.9545364891518737

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 0.1 batch size: 16 best_epoch: 18 score: 0.9643984220907298

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 0.1 batch size: 32 best_epoch: 8 score: 0.9683979837825991

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 0.1 batch size: 64 best_epoch: 3 score: 0.9640368178829718

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 0.5 batch size: 16 best_epoch: 18 score: 0.9655051501205347

Jack Vacek
Jaesun Park

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 0.5 batch size: 32 best_epoch: 18 score: 0.9611330265176419

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 0.5 batch size: 64 best_epoch: 10 score: 0.9639710716633793

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 1.0 batch size: 16 best_epoch: 5 score: 0.9519614288845059

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 1.0 batch size: 32 best_epoch: 18 score: 0.9583497698882314

Using downloaded and verified file: ./med/pneumoniamnist.npz
Using downloaded and verified file: ./med/pneumoniamnist.npz
model_name: googlenet
for model: googlenet lr: 1.0 batch size: 64 best_epoch: 17 score: 0.9675213675213675

With this output, we are able to determine between each model and their default parameters of a learning rate of 0.1 and batch size of 32 which model performs the best with this table below:

	ResNet18	ResNet101	GoogleNet
BreastMNIST Score	0.706140350877193	0.5864661654135339	0.8830409356725145
PneumoniaMNIST Score	0.9648805610344072	0.9071115494192417	0.9683979837825991

This shows that the GoogleNet model type performs best given the default parameters of the program. However, what about the best score overall? That is given in the table below:

	ResNet18	ResNet101	GoogleNet
BreastMNIST Score	0.8460735171261486	0.6785714285714286	0.8830409356725145
PneumoniaMNIST Score	0.9700964277887355	0.9599386368617138	0.9683979837825991

This shows that GoogleNet performs best on the BreastMNIST dataset but that the ResNet18 performs best on the PneumoniaMNIST set when optimizing each of the model's optimizer parameters between the available choices. This suggests that while the model does impact the performance of the accuracy, so does changing the optimizer parameters. However it is hard to tell how each of these parameters changes the accuracy.

We initially perform an experiment using different neural networks and different learning rate parameters. We trained and evaluated 5 different learning rate parameters - 0.01, 0.1, 0.5, 1.0 - each going through 20 epochs, where we also selected the “best” epoch that performed the best from the rest. Below are the performance results of each neural network, given the learning rate and the best epoch where it performed the best.

Scores of the **ResNet18** neural network architecture of **BreastMNIST**

	Learning Rate	Best Epoch	Score
ResNet18 (breastmnist)	0.01	16	0.7502088554720133
ResNet18 (breastmnist)	0.1	16	0.706140350877193
ResNet18 (breastmnist)	0.5	16	0.5668337510442774
ResNet18 (breastmnist)	1.0	6	0.4578111946532999

We can see that ResNet18 continues to decrease as the learning rate increases.

Scores of the **GoogleNet** neural network architecture of **BreastMNIST**

	Learning Rate	Best Epoch	Score
GoogleNet (breastmnist)	0.01	9	0.815998329156224
GoogleNet (breastmnist)	0.1	13	0.8830409356725145
GoogleNet (breastmnist)	0.5	18	0.7715121136173768
GoogleNet (breastmnist)	1.0	19	0.6593567251461989

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While GoogleNet increases the score initially, as the learning rate continues to increase, the score gradually decreases.

Scores of the **ResNet101** neural network architecture of **BreastMNIST**

	Learning Rate	Best Epoch	Score
ResNet101 (breastmnist)	0.01	9	0.5952380952380952
ResNet101 (breastmnist)	0.1	11	0.5864661654135339
ResNet101 (breastmnist)	0.5	18	0.6063074352548037
ResNet101 (breastmnist)	1.0	15	0.6558061821219716

ResNet101 sees a positive trend whereas the learning rate increases, the score also continues to increase.

Scores of the **ResNet18** neural network architecture of **PneumoniaMNIST**

	Learning Rate	Best Epoch	Score
ResNet18 (pneumoniamnist)	0.01	10	0.9504163927240851
ResNet18 (pneumoniamnist)	0.1	9	0.9648805610344072
ResNet18 (pneumoniamnist)	0.5	16	0.9672145518299364
ResNet18 (pneumoniamnist)	1.0	5	0.9649901380670611

Similar to the ResNet18 model in BreastMnist, as the learning rate increases, we can see that there's a slight drop from 0.5 to 1.0. This means if the learning rate were to continue to increase, then the overall score would most likely continue to decrease.

Jack Vacek
Jaesun Park

Scores of the **GoogleNet** neural network architecture of **PneumoniaMNIST**

	Learning Rate	Best Epoch	Score
GoogleNet (pneumoniamnist)	0.01	15	0.9627328511943897
GoogleNet (pneumoniamnist)	0.1	8	0.9683979837825991
GoogleNet (pneumoniamnist)	0.5	18	0.9611330265176419
GoogleNet (pneumoniamnist)	1.0	18	0.9583497698882314

Similar to the analysis on ResNet18 for PneumoniaMNIST, GoogleNet sees a similar trend where it decreases overall as the learning rate increases.

Scores of the **ResNet101** neural network architecture of **PneumoniaMNIST**

	Learning Rate	Best Epoch	Score
ResNet101 (pneumoniamnist)	0.01	15	0.9275367083059392
ResNet101 (pneumoniamnist)	0.1	19	0.9071115494192417
ResNet101 (pneumoniamnist)	0.5	8	0.9435349550734166
ResNet101 (pneumoniamnist)	1.0	13	0.9436664475126014

The ResNet101 experiences an overall increase, similar to the ResNet101 model on BreastMNIST.

Based on the results, we make a deduction that while ResNet101 performed better as the learning rate increases, ResNet18 and GoogleNet actually performed worse as the learning rate increases.

Alternatively, we do the same experiment with batch size, varying in different sizes given the default learning rate of 0.1.

Jack Vacek
Jaesun Park

Scores of the **ResNet18** neural network architecture of **BreastMNIST**

	Batch Size	Best Epoch	Score
ResNet18 (breastmnist)	16	19	0.7412280701754386
ResNet18 (breastmnist)	32	16	0.706140350877193
ResNet18 (breastmnist)	64	19	0.8460735171261486

Here, we see fluctuations with the score when the batch size goes from 16 to 32 to 64.

Scores of the **GoogleNet** neural network architecture of **BreastMNIST**

	Batch Size	Best Epoch	Score
GoogleNet (breastmnist)	16	19	0.8771929824561403
GoogleNet (breastmnist)	32	13	0.8830409356725145
GoogleNet (breastmnist)	64	9	0.8613199665831245

Similar to the previous results, GoogleNet also fluctuates, where it decreases in score when going from 32 to 64.

Scores of the **ResNet101** neural network architecture of **BreastMNIST**

	Batch Size	Best Epoch	Score
ResNet101 (breastmnist)	16	14	0.599624060150376
ResNet101 (breastmnist)	32	11	0.5864661654135339
ResNet101 (breastmnist)	64	2	0.5653717627401837

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Jaesun Park

For the ResNet101 neural network, the overall score decreases as the batch size increases.

Scores of the **ResNet18** neural network architecture of **PneumoniaMNIST**

	Batch Size	Best Epoch	Score
ResNet18 (pneumoniamnist)	16	14	0.9700964277887355
ResNet18 (pneumoniamnist)	32	9	0.9648805610344072
ResNet18 (pneumoniamnist)	64	6	0.955698005698006

In PenumoniaMNIST, the ResNet18 performs worse as the batch size increases.

Scores of the **GoogleNet** neural network architecture of **PneumoniaMNIST**

	Batch Size	Best Epoch	Score
GoogleNet (pneumoniamnist)	16	18	0.9643984220907298
GoogleNet (pneumoniamnist)	32	8	0.9683979837825991
GoogleNet (pneumoniamnist)	64	3	0.9640368178829718

Using the GoogleNet neural network, the score fluctuates as it increases then decreases as it goes from 16 to 32 to 64

Scores of the **ResNet101** neural network architecture of **PneumoniaMNIST**

	Batch Size	Best Epoch	Score
ResNet101 (pneumoniamnist)	16	18	0.9599386368617138
ResNet101 (pneumoniamnist)	32	19	0.9071115494192417
ResNet101	64	19	0.2648367302213456

Jack Vacek
Jaesun Park

(pneumoniamnist)			

In the ResNet101 neural network, there's a drastic decrease in the score as the batch size reaches 64. In this case, the best score is actually when the batch size is set to 16.

Overall, we can see a clear correlation with the increase in batch size and its score. As the batch size increases to 64, the model actually performs worse than when it's around 16-32. From this analysis, we can deduce that an overly increasing batch size can detrimentally affect the score.